Primitive subgroup pairs from tables of marks

We complete the counting and suborbit comparisons left open by the previous character calculation for Leemans and Rodrigues, *On primitive geometries of rank two*, Art Discrete Appl. Math. 2 (2019), P1.08. The fixed source is the twelve-page journal PDF. Our conclusions concern its Tables 1, 2 and 5–8. They do not establish its individual M_{11} geometries or designs in Tables 3–4. The earlier Rudvalis correction is unchanged.

The mathematical input is the accompanying, previously reviewed note *Counting maximal-subgroup pairs from fixed points*, retained exactly as `theory.tex`. Its normalizer lemma, Burnside formula and product of marks theorem are used here with their stated hypotheses. The present application uses complete subgroup tables from TomLib 1.2.11 and compatible element data from the installed Character Table Library. The classical identification of the named groups, their automorphism groups, and their maximal subgroup shapes is borrowed. These are premises, not classifications proved by consistency tests on the tables.

1 Full automorphism equivalence

Let G be one of $M_{12}, J_2, J_3, HS, McL, He$, and put $A = \mathrm{Aut}(G) = G.2$. Write H_i for the subgroup representatives supplied by the complete mark table of A, and

$$m_{ij} = |(A/H_i)^{H_j}|.$$

The unique class of order $|A|/2$ is G. The maximal proper classes below it give all A-classes of maximal subgroups of G. If M is one of them, then $[N_A(M) : M] = m_{MM}$ is one or two. In the latter case, the unique containing class of order $2|M|$ is its normalizer, as proved in the normalizer lemma. No subgroup enumeration is required at this stage.

Let $\psi(a)$ be the sum of the normalizer mark rows at the cyclic class $\langle a \rangle$. This is the number of maximal subgroups of G fixed by $a \in A$. Put

$$r = \sum_{[a]_A} \frac{\psi(a)}{|C_A(a)|}, \quad s = \sum_{[a]_A} \frac{\psi(a)^2}{|C_A(a)|}, \quad t = \sum_{[a]_A} \frac{\psi(a^2)}{|C_A(a)|}.$$

Here r counts maximal subgroup classes under A, s counts ordered pairs, and $(s - 2r + t)/2$ counts unordered pairs of distinct subgroups. The exact arithmetic gives the following values.

G	all maximal subgroups	r	s	t	unordered pairs
M_{12}	3446	8	288	60	166
J_2	16964	9	585	101	334
J_3	192358	8	975	133	546
HS	79962	10	595	103	339
McL	544129	10	799	107	443
He	6920939	10	7643	525	4074

Theorem 1. *Under the stated library and named-group premises, these six numbers are the numbers of primitive rank-two coset geometries in the source's* $\mathrm{Aut}(G)$ *equivalence. They agree with all six corresponding entries of its Table 1.*

Proof. The source's equivalence is the action of $\mathrm{Aut}(G)$ on unordered pairs of distinct maximal subgroups. The accepted coset-geometry argument identifies these pairs with its firm, connected primitive geometries. The normalizer lemma identifies every transitive component of the action on maximal subgroups. The stored cyclic fusion turns its marks into the fixed-point values ψ. For each element, a fixed unordered pair is either two fixed points or a two-cycle, giving $(\psi(a)^2-2\psi(a)+\psi(a^2))/2$ fixed pairs. Summing over element classes gives the displayed integer. The complete input rows, all selected classes, fixed-point arrays and rational sums are retained in the certificate; the checker recomputes them from the literal input without the source totals as inputs. □

Together with the accepted character totals and the imported M_{22} census, this accounts for every Table 1 count: eleven agree, while the already corrected Rudvalis value is 556287 rather than the printed 1511. This statement adds no second refutation credit.

2 Suborbit multisets and the preliminary counts

For maximal subgroups $H, K \leq G$, decompose the pointwise product of their mark rows as

$$m_H m_K = \sum_L c_L m_L.$$

The product theorem gives c_L orbits of length $|H|/|L|$ for H on G/K. The representative L need only be conjugate to an intersection $H \cap K^g$; it need not itself lie in the fixed representative H.

The retained sparse table is triangular. Starting with the largest class position, the checker divides the remaining coordinate by its diagonal mark, records c_L, and subtracts $c_L m_L$. It verifies integrality and nonnegativity at every step and checks that the final residual is zero. It then checks that the lengths sum to $[G : K]$. A singleton occurs exactly for equal maximal classes. Thus every printed multiset has an explicit nonnegative decomposition certificate, not just a matching total size.

G	source table	cells	possible degree joins	changed numeric cells
M_{11}	2	25	1	0
M_{12}	5	121	16	0
M_{22}	6	64	2	0
M_{23}	7	49	2	0
J_1	8	49	1	0

All 308 numerical cells agree, with the M_{23} label correction proved below. Equal subgroup orders are handled by exhausting the simultaneous row and column bijections allowed by equal indices. The retained joins specify actual mark-table class numbers. Superscripts in the source cells are multiplicities; their extraction is bound to the fixed PDF and checked against its native pages. Numerical agreement does not by itself identify the subgroup labels.

The upper triangular counts in the source are a different intermediate quantity from unordered-pair orbits. For each two distinct maximal classes, one orientation contributes all double cosets. On the diagonal one removes the singleton but initially keeps both double cosets when inversion interchanges them. Thus the upper count can exceed the final inner count. The exact products and the separate Burnside calculation give

	upper triangle, no singletons	inner unordered pairs	full automorphism pairs
M_{11}	38	37	37
M_{12}	268	255	166.

These recover the printed preliminary values 38 and 268 and explain why neither should be compared directly with the final 166 for M_{12}. Since inversion is an involution on the diagonal double cosets, the decrease from 38 to 37 for M_{11} means that exactly one pair of distinct diagonal orbits is identified by duality, as the source states. The mark calculations also reproduce the previously accepted totals $37, 170, 669$ for M_{11}, M_{23}, J_1. The value 104 obtained with the M_{22} mark table is an inner-automorphism count; its full-automorphism value 81 is imported from the accepted L073 census.

3 Distinguishing the two M_{23} labels

The two maximal subgroups of index 253 have the classical shapes $L_3(4) : 2$ and $2^4 : A_7$. Their permutation characters coincide. Therefore the character-library convention for ordering primitive characters does not, by itself, distinguish these two embeddings.

Lemma 2. *The number of conjugates of $L \leq G$ contained in $M \leq G$ is*

$$m_{M,L} \frac{|M|}{|N_G(L)|}.$$

In particular, if this number is one, that conjugate of L is normal in M.

Proof. Count the elements $g \in G$ with $g^{-1}Lg \leq M$. Dividing their number by $|M|$ gives the fixed cosets counted by $m_{M,L}$. Dividing it by $|N_G(L)|$ counts distinct conjugates of L in M. Conjugation by M permutes these subgroups, so a unique one is normal. □

In the acquired table of M_{23}, take M in class 201 and L in class 33. The exact entries are

$$|M| = 40320, \qquad |L| = 16, \qquad m_{M,L} = 1, \qquad m_{L,L} = 2520.$$

Consequently $|N_G(L)| = 16 \cdot 2520 = 40320$, and the lemma gives exactly one conjugate of L in M. The complete cyclic fusion shows that the only cyclic classes contained in L are classes 1 and 2, of orders 1 and 2. Equivalently, the contained order-two subgroup counts sum to 15. Every nonidentity element of L is therefore an involution. Such a group is abelian, since $(xy)^{-1} = xy = y^{-1}x^{-1} = yx$, and hence $L \cong C_2^4$.

The group $L_3(4) : 2$ cannot contain a normal subgroup of order 16. Indeed, its intersection with the nonabelian simple subgroup $L_3(4)$ would be trivial, so it would inject into the quotient of order two. The borrowed list of the two maximal shapes therefore identifies class 201 as $2^4 : A_7$ and class 202 as $L_3(4) : 2$.

The native source Table 7 labels its second row $L_3(4) : 2$ and gives orbits $7, 16$ on the 23 points. That row is class 201, so its label must be $2^4 : A_7$. Its third row, printed $2^4 : A_7$, has orbits $2, 21$ and is class 202; its label must be $L_3(4) : 2$. Exchanging these two row labels corrects the table. The two column headings are both 253; the explicit class join specifies their interpretation. No numerical cell is changed. This is a source-label correction, separate from the Rudvalis counting error and from any transcription error in our work.

4 Certificate and attribution

One bounded GAP call exported stored data only; it built no groups, cosets or geometries and performed no fusion search. The raw output and receipt are retained. GAP printed four identity lists as intervals $[1 .. n]$. The parser expands precisely that literal notation outside quoted strings; it does not change any mathematical value. The two A_7 classes of M_{22} legitimately share a character-table name, so class positions, not distinct names, are retained.

The numerical data and the original counts are attributed to the source and the GAP libraries as appropriate. Burnside's lemma, tables of marks, their product formula and elementary transporter counting are classical. Our contribution here is the common calculation with full automorphism equivalence, the explicit source joins and the certified label correction. Library completeness, named-group identification and compatible fusion remain visible assumptions. There is no new Lean formalization, measured compression factor, or whole-source admission in this component.

5 Exact class joins

For reference, the following are the mark-table class numbers in source row order. The same lists identify the columns. For M_{23}, use the corrected row labels just proved. The order of the acquired mark table is part of the exact input; these numbers are not identifiers independent of that table edition.

G	mark classes in source order
M_{11}	$38, 37, 35, 34, 27$
M_{12}	$146, 145, 144, 143, 135, 134, 133, 130, 126, 127, 113$
M_{22}	$155, 154, 153, 152, 151, 150, 146, 145$
M_{23}	$203, 201, 202, 200, 198, 196, 156$
J_1	$39, 38, 37, 36, 35, 34, 28$

The exhaustive degree joins have two exact solutions for M_{12} and M_{22} and one for each other group. All solutions are retained, rather than selecting one to hide a discrepancy. For M_{11}, M_{12}, J_1, the maximal permutation characters are pairwise different, so the documented character ordering distinguishes their named classes. The repeated M_{22} character occurs for the two classes of the same shape A_7. For M_{23}, where the repeated character belongs to different shapes, the normal subgroup argument supplies the additional identification. The complete lists of coefficients c_L, both joins where applicable, and every printed cell are retained in `results.json` and `source_join.json`.